XISOSIS · Institutional Research Paper

THE ALGEBRAIC FORMULATION OF ℚ-NUMBERS IN QUANTUM INFORMATION SYSTEMS AND MITIGATION OF THE BARREN PLATEAU PHENOMENON

Main result: every ℚ-structured circuit has its dynamical Lie algebra inside $\mathfrak{sp}(2^{n-1})$ or $\mathfrak{so}(2^n)$, and its loss variance is exactly $1/(2^n - (-1)^n)$, the Haar value up to ±1; the quaternionic restriction alone therefore does not remove the barren plateau.

Roger Cooperman
https://orcid.org/0009-0009-5011-2135
https://doi.org/10.5281/zenodo.23361358
and A.E. Petrunina V.V.
https://orcid.org/0009-0002-4595-736X

Vienna · 12 October 2026

© 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License (see Licence)

Abstract — Variational quantum algorithms (VQAs) suffer from barren plateaus: for sufficiently expressive parameterized quantum circuits (PQCs) the variance of the cost function and of its gradients decays exponentially in the number of qubits n. The Lie-algebraic theory of barren plateaus identifies the dimension of the circuit's dynamical Lie algebra (DLA) as the controlling quantity. We study circuits built from *Qubit Numbers* $\mathbb{Q}$, the real algebra spanned by $\{1, i, \mathfrak{q}, i\mathfrak{q}\}$, which is isomorphic to Hamilton's quaternions $\mathbb{H}$ and whose unit elements form $SU(2)$. We make the induced n-qubit structure precise: a $\mathbb{Q}$-structured circuit is one that commutes with the antiunitary operator $J_n = \sigma_y^{\otimes n} K$, which satisfies $J_n^2 = (-1)^n$. We prove that a Pauli rotation is $\mathbb{Q}$-structured if and only if its Pauli string has odd weight, and that the DLA of any such circuit is contained in $\mathfrak{sp}(2^{n-1})$ for odd n and $\mathfrak{so}(2^n)$ for even n, with dimension $(4^n - (-2)^n)/2$. A natural entangling $\mathbb{Q}$-ansatz saturates this bound (verified exactly for $n \leq 9$). For it, the loss variance is $1/(2^n - (-1)^n)$, which is the Haar value up to ± 1 in the denominator: **the $\mathbb{Q}$-restriction by itself does not mitigate barren plateaus.** Conversely, we exhibit a $\mathbb{Q}$-structured "cluster" ansatz whose DLA is $\mathfrak{so}(2\lceil n/2 \rceil) \oplus \mathfrak{so}(2\lfloor n/2 \rfloor)$ (dimension $\Theta(n^2)$) and whose loss variance is $1/(2\lceil n/2 \rceil - 1)$. This family is trainable, but it is free-fermionic and therefore efficiently classically simulable. All dimension and variance statements are checked by exact Lie-closure computations and state-vector simulations. This revision corrects several claims in an earlier version of the manuscript; Appendix A lists each change. See Appendix B for Validation Code access.

1. Introduction

Variational quantum algorithms and quantum neural networks train a parameterized unitary $U(\boldsymbol{\theta})$ to minimize a loss $\ell(\boldsymbol{\theta}) = \mathrm{Tr}[O\, U(\boldsymbol{\theta})\rho\, U(\boldsymbol{\theta})^\dagger]$. McClean *et al.* [1] showed that when $U(\boldsymbol{\theta})$ approximates a unitary 2-design over $\mathcal{U}(2^n)$, $\mathrm{Var}_{\boldsymbol{\theta}}[\partial_\theta \ell] \in \mathcal{O}(2^{-n})$ — the barren plateau (BP). Subsequent work tied BPs to cost locality [9], expressibility [10], local minima [8] and, decisively, to the dynamical Lie algebra

$$\mathfrak{g} = \langle iH_1, \ldots, iH_K \rangle_{\mathrm{Lie}},$$

generated by the circuit's generators [2]; DLAs of common spin-model ansätze have since been classified [7]. For deep circuits the loss variance is given exactly by the adjoint-representation formula of Ragone *et al.* [3] and Fontana *et al.* [4]: for simple $\mathfrak{g}$ and $O \in i\mathfrak{g}$,

$$\mathrm{Var}_{\boldsymbol{\theta}}[\ell] = \frac{\mathcal{P}_{\mathfrak{g}}(\rho)\, \mathcal{P}_{\mathfrak{g}}(O)}{\dim \mathfrak{g}}, \qquad \mathcal{P}_{\mathfrak{g}}(H) = \sum_j \left|\mathrm{Tr}\left(B_j^\dagger H\right)\right|^2,$$

where $\{B_j\}$ is a Hilbert–Schmidt-orthonormal basis of $\mathfrak{g}$. Polynomial variance therefore needs either a polynomially small $\mathfrak{g}$ or exponentially concentrated projections $\mathcal{P}_{\mathfrak{g}}$. Cerezo *et al.* [5] argue that the structures that provably avoid BPs typically also make the model classically simulable.

This paper. We ask what the quaternionic algebra $\mathbb{Q} \cong \mathbb{H}$ does to the DLA. Quaternions are not new to quantum information. The isomorphism of unit quaternions with $SU(2)$ is

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

classical, quaternionic quantum mechanics is a mature subject [14, 16], unit quaternions are already used to parameterize single-qubit gates in sequential VQA optimizers such as Free Quaternion Selection [15], and in that quaternion parameterization the concentration (flatness) of the landscape has been analyzed with unitary-design techniques [19]. Writing a qubit's amplitude pair as a single quaternion is standard in the Hopf-fibration geometry of qubits [20]. To our knowledge, the *dynamical-Lie-algebra* consequences of the quaternionic structure, which are the subject of this paper, have not been worked out explicitly. Our contribution is an exact account of the n-qubit symmetry a $\mathbb{Q}$-structure imposes on a PQC and of its consequences for trainability:

1. A precise definition of $\mathbb{Q}$-structured circuits via the antiunitary J_n (Section 3) and a proof that $\mathbb{Q}$-structured Pauli rotations are exactly the odd-weight ones.
2. The exact DLA bound $\mathfrak{g} \subseteq \mathfrak{sp}(2^{n-1})$ or $\mathfrak{so}(2^n)$, with saturation by an entangling ansatz (Section 3).
3. The exact loss variance of that ansatz, which shows the BP persists at the Haar rate (Section 4).
4. A $\mathbb{Q}$-structured ansatz with $\Theta(n^2)$-dimensional DLA and $\Theta(1/n)$ variance, together with a proof that it is classically simulable (Section 5).
5. Numerical confirmation of every dimension and variance claim (Section 6), with open validation code (Appendix B).

2. The ℚ-Number Algebra

Motivation: the quantum unit $\mathfrak{q}$**.** A qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$, is a pair of complex numbers. To collapse this pair into a single "number," introduce a new unit $\mathfrak{q}$ that marks the $|1\rangle$ amplitude, while the complex part carries the $|0\rangle$ amplitude:

$$Z = \alpha + \mathfrak{q}\beta.$$

For this to behave as a number system, $\mathfrak{q}$ must interact consistently with the imaginary unit i. We take $\mathfrak{q}$ to be a second imaginary unit that anticommutes with i, so that i, $\mathfrak{q}$ and their product play the roles of Hamilton's i, j, k. This is the Cayley–Dickson doubling $\mathbb{H} = \mathbb{C} \oplus \mathfrak{q}\,\mathbb{C}$. Writing qubit amplitudes as quaternions in this way is also used in the Hopf-fibration geometry of qubits [20].

Ordering matters. Because $\mathfrak{q}$ does not commute with i, $\mathfrak{q}\beta \neq \beta\mathfrak{q}$ in general: with $\beta = c + di$, $\mathfrak{q}\beta = c\mathfrak{q} + d\,\mathfrak{q}i$ while $\beta\mathfrak{q} = c\mathfrak{q} + d\,i\mathfrak{q} = c\mathfrak{q} - d\,\mathfrak{q}i$. We place $\mathfrak{q}$ on the left. This is the ordering for which the matrix representation below sends Z to the state $\alpha|0\rangle + \beta|1\rangle$ (Proposition 1). With $\beta\mathfrak{q}$ the same representation would produce $\alpha|0\rangle + \bar{\beta}|1\rangle$ instead.

Scope. The "single number per state" picture holds for one qubit only. For $n \geq 2$ qubits, $\mathbb{Q}^{\otimes n}$ is not a division algebra ($\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$), and two-qubit states already need a pair of quaternions [20]. Section 3 gives the correct n-qubit extension, which is a symmetry rather than a number system.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

Definition 1. $\mathbb{Q}$ is the four-dimensional real associative algebra generated by i and $\mathfrak{q}$ with relations

$$i^2 = \mathfrak{q}^2 = -1, \qquad i\mathfrak{q} = -\mathfrak{q}i.$$

With $\alpha = a + bi$ and $\beta = c + di$, a general element is

$$Z = \alpha + \mathfrak{q}\beta = a + bi + c\mathfrak{q} + d(\mathfrak{q}i), \qquad a, b, c, d \in \mathbb{R},$$

equivalently $Z = a + bi + c\mathfrak{q} - d(i\mathfrak{q})$, since $\mathfrak{q}i = -i\mathfrak{q}$. The conjugate is $Z^* = \bar{\alpha} - \mathfrak{q}\beta = a - bi - c\mathfrak{q} - d(\mathfrak{q}i)$. Setting $j := \mathfrak{q}$ and $k := i\mathfrak{q}$ recovers Hamilton's relations $i^2 = j^2 = k^2 = ijk = -1$, so $\mathbb{Q} \cong \mathbb{H}$.

Representation. Define the real-linear map $\phi: \mathbb{Q} \to M_2(\mathbb{C})$ by

$$\phi(1) = I, \quad \phi(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_z, \quad \phi(\mathfrak{q}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y, \quad \phi(\mathfrak{q}i) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_x.$$

Here $\phi(\mathfrak{q})$ sends $|0\rangle \mapsto |1\rangle$, which is the placeholder role of $\mathfrak{q}$, and

$$\phi(Z) = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} = \begin{pmatrix} a + bi & -c + di \\ c + di & a - bi \end{pmatrix}.$$

Lemma 1 (multiplication table). $(i\mathfrak{q})^2 = (\mathfrak{q}i)^2 = -1$ and $i\mathfrak{q}\mathfrak{q}i = 1$; ϕ is an injective algebra homomorphism.

Proof. $(i\mathfrak{q})^2 = i\mathfrak{q}i\mathfrak{q} = -ii\mathfrak{q}\mathfrak{q} = -(-1)(-1) = -1$, and likewise for $\mathfrak{q}i = -i\mathfrak{q}$. Also $i\mathfrak{q}\mathfrak{q}i = i(\mathfrak{q}^2)i = -i^2 = 1$. The matrices $i\sigma_z$ and $-i\sigma_y$ square to $-I$ and anticommute, and $\phi(\mathfrak{q})\phi(i) = (-i\sigma_y)(i\sigma_z) = \sigma_y\sigma_z = i\sigma_x = \phi(\mathfrak{q}i)$, so ϕ respects all relations. Injectivity follows from the linear independence of $\{I, \sigma_x, \sigma_y, \sigma_z\}$. ■

Lemma 2 (norm = determinant). $ZZ^* = |\alpha|^2 + |\beta|^2 = a^2 + b^2 + c^2 + d^2 = \det\phi(Z)$.

Proof. $\det\phi(Z) = \alpha\bar{\alpha} - (-\bar{\beta})\beta = |\alpha|^2 + |\beta|^2$. The product ZZ^* is computed directly from the relations. ■

Lemma 3 (trace orthogonality). For $e_\mu, e_\nu \in \{1, i, \mathfrak{q}, \mathfrak{q}i\}$, $\mathrm{Tr}\left(\phi(e_\mu)^\dagger \phi(e_\nu)\right) = 2\delta_{\mu\nu}$.

Proof. Up to phases of modulus one, $\phi(e_\mu)$ runs over the Pauli basis $\{I, \sigma_z, \sigma_y, \sigma_x\}$, and $\mathrm{Tr}(\sigma_\alpha\sigma_\beta) = 2\delta_{\alpha\beta}$. ■

Lemma 4 (imaginary part is $\mathfrak{su}(2)$). $[i, \mathfrak{q}] = 2i\mathfrak{q}$, $[\mathfrak{q}, i\mathfrak{q}] = 2i$, $[i\mathfrak{q}, i] = 2\mathfrak{q}$, and the same brackets hold for their images under ϕ. Hence $\phi(\mathrm{Im}\,\mathbb{Q}) = \mathfrak{su}(2)$.

Proof. By Lemma 1, $\mathfrak{q}i = -i\mathfrak{q}$, so $[i, \mathfrak{q}] = 2i\mathfrak{q}$. Next, $\mathfrak{q}(i\mathfrak{q}) = -i\mathfrak{q}^2 = i$ and $(i\mathfrak{q})\mathfrak{q} = -i$ give $[\mathfrak{q}, i\mathfrak{q}] = 2i$. The third bracket follows in the same way. ϕ is a homomorphism, so the matrix brackets agree. ■

Lemma 5 (unit $\mathbb{Q}$-numbers). ϕ maps $\{Z: ZZ^* = 1\} \cong S^3$ isomorphically onto $SU(2)$.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

Proof. $\phi(Z)^\dagger = \phi(Z^*)$, so $\phi(Z)^\dagger\phi(Z) = \phi(Z^*Z) = I$, and $\det\phi(Z) = 1$ by Lemma 2. Every $SU(2)$ matrix has the form $\begin{pmatrix}\alpha & -\bar{\beta}\\ \beta & \bar{\alpha}\end{pmatrix}$ with $|\alpha|^2 + |\beta|^2 = 1$, which is $\phi(\alpha + \mathfrak{q}\beta)$. ■

Remark (states vs. operators). A ℚ-number is primarily an *operator*: a gate in $SU(2)$. The motivating state $\alpha|0\rangle + \beta|1\rangle$ is recovered as the first column of that gate, $|\psi_Z\rangle := \phi(Z)|0\rangle$. We use Z as a gate throughout and obtain states only through this relation. Mixed single-qubit states are density matrices, which form the Bloch ball, and are not ℚ-numbers.

Proposition 1 (state correspondence). For unit $Z = \alpha + \mathfrak{q}\beta$, $\phi(Z)|0\rangle = \alpha|0\rangle + \beta|1\rangle$, so every pure state arises this way. The choice of Z is unique only up to $Z \mapsto Z\,e^{\gamma i}$, $\gamma \in \mathbb{R}$, which multiplies the state by the global phase $e^{i\gamma}$. The state space is therefore $S^3/S^1 \cong S^2$, the Hopf fibration [20].

Proof. The first column of $\phi(Z)$ is $(\alpha, \beta)^\mathsf{T}$, which reaches every unit vector of $\mathbb{C}^2$. Moreover, $Ze^{\gamma i} = \alpha e^{i\gamma} + \mathfrak{q}\left(\beta e^{i\gamma}\right)$, because $e^{\gamma i}$ commutes with α and β. So right multiplication by $e^{\gamma i}$ maps the state to $e^{i\gamma}(\alpha|0\rangle + \beta|1\rangle)$. ■

3. ℚ-Structure on n Qubits and the Exact DLA

Single qubit. Let $J_1 = \sigma_y K$, where K is complex conjugation. A direct computation shows that $M \in M_2(\mathbb{C})$ satisfies $J_1 M J_1^{-1} = \sigma_y \bar{M} \sigma_y = M$ exactly when $M = \begin{pmatrix}\alpha & -\bar{\beta}\\ \beta & \bar{\alpha}\end{pmatrix}$, that is, when $M = \phi(Z)$ for some $Z \in \mathbb{Q}$. So $\phi(\mathbb{Q})$ is *precisely* the commutant of the antiunitary J_1, and $J_1^2 = \sigma_y\bar{\sigma}_y = -I$ is the quaternionic case of Dyson's threefold way [13]. This gives the canonical n-qubit extension.

Definition 2. $J_n := J_1^{\otimes n} = \sigma_y^{\otimes n} K$. A unitary U on $(\mathbb{C}^2)^{\otimes n}$ is ℚ-*structured* if $J_n U J_n^{-1} = U$. A PQC is ℚ-structured if every gate is.

Lemma 6. $J_n^2 = (-1)^n I$.

Proof. $J_n^2 = \sigma_y^{\otimes n}\,\overline{\sigma_y}^{\otimes n} = \left(-\sigma_y^2\right)^{\otimes n} = (-1)^n I$. ■

The n-fold tensor product of the quaternionic structure is therefore quaternionic only for odd n; for even n it is a *real* structure. This matches the algebra isomorphism $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$.

Proposition 2 (odd-weight criterion). Let P be an n-qubit Pauli string of weight $w(P)$, the number of non-identity factors. Then $e^{-i\theta P/2}$ is ℚ-structured for all θ if and only if $w(P)$ is odd.

Proof. For a single qubit, $\sigma_y\bar{\sigma}_\alpha\sigma_y = -\sigma_\alpha$ for $\alpha \in \{x, y, z\}$ and $\sigma_y\bar{I}\sigma_y = I$. Hence $J_n P J_n^{-1} = (-1)^{w(P)}P$. Because J_n is antilinear, $J_n(-iP)J_n^{-1} = +i(-1)^{w(P)}P$, which equals $-iP$ if and only if $w(P)$ is odd. ■

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

In particular, all single-qubit rotations are ℚ-structured. Every two-qubit Pauli rotation (ZZ, XX, …) **breaks** the structure. The lowest-weight ℚ-structured entanglers are three-body, such as $e^{-i\theta Z_j Z_{j+1} Z_{j+2}/2}$.

Theorem 1 (DLA bound). Let $\mathfrak{k}_n := \{A \in \mathfrak{su}(2^n) : J_n A J_n^{-1} = A\}$. Then

$$\mathfrak{k}_n \cong \begin{cases} \mathfrak{sp}(2^{n-1}), & n \text{ odd}, \\ \mathfrak{so}(2^n), & n \text{ even}, \end{cases} \qquad \dim \mathfrak{k}_n = \frac{4^n - (-2)^n}{2},$$

and the DLA of every ℚ-structured PQC satisfies $\mathfrak{g} \subseteq \mathfrak{k}_n$.

Proof. The unitaries commuting with an antiunitary J with $J^2 = -1$ (resp. $+1$) on $\mathbb{C}^N$ form $Sp(N/2)$ (resp. a conjugate of $O(N)$) [13]. Taking $N = 2^n$ and applying Lemma 6 identifies $\mathfrak{k}_n$. By Proposition 2, $\mathfrak{k}_n$ is spanned by $\{iP : w(P) \text{ odd}\}$. The number of odd-weight strings is $\sum_{w \text{ odd}} \binom{n}{w} 3^w = (1+3)^n - (1-3)^n/2$, which matches $\dim \mathfrak{sp}(2^{n-1}) = 2^{n-1}(2^n + 1)$ for odd n and $\dim \mathfrak{so}(2^n) = 2^{n-1}(2^n - 1)$ for even n. Generators lie in $\mathfrak{k}_n$, and $\mathfrak{k}_n$ is closed under brackets, so $\mathfrak{g} \subseteq \mathfrak{k}_n$. ■

Corollary 1. $\dim \mathfrak{k}_n / \dim \mathfrak{su}(2^n) \to 1/2$. The ℚ-restriction removes about half of $\mathfrak{su}(2^n)$, and $\dim \mathfrak{k}_n \in \Theta(4^n)$ is exponential, not polynomial.

Proposition 3 (saturation; computer-verified). The entangling ℚ-ansatz with generators

$$\mathcal{G}_{\text{ent}} = \{X_j, Z_j\}_{j=1}^{n} \cup \{Z_j Z_{j+1} Z_{j+2}\}_{j=1}^{n-2}$$

has $\mathfrak{g} = \mathfrak{k}_n$ for $2 \le n \le 9$, verified by exact Pauli-string Lie closure (Table 1). For $n = 2$ the bound is met by local gates alone, since $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$. We conjecture saturation for all $n \ge 2$.

4. Gradient Variance of Generic ℚ-Circuits

Theorem 2 (barren plateau persists). Let U be a deep ℚ-structured PQC with $\mathfrak{g} = \mathfrak{k}_n$, $n \ge 3$ (so $\mathfrak{k}_n$ is simple), let $\rho = |+\rangle\langle+|^{\otimes n}$ and $O = X_1$. Then

$$\mathrm{Var}_{\boldsymbol{\theta}}[\ell] = \frac{1}{2^n - (-1)^n} = \begin{cases} \dfrac{1}{2^n + 1}, & n \text{ odd}, \\ \dfrac{1}{2^n - 1}, & n \text{ even}. \end{cases}$$

For comparison, a Haar-random circuit gives $\mathrm{Var}_{\mathrm{Haar}}[\ell] = 1/(2^n + 1)$.

Proof. Use (1) with orthonormal basis $\{P/\sqrt{2^n} : w(P) \text{ odd}\}$. Here $\rho = 2^{-n} \sum_{P \in \{I,X\}^{\otimes n}} P$, so $\mathrm{Tr}(P\rho) = 1$ for the 2^{n-1} odd-weight X-strings and 0 for all other odd-weight strings. This gives $\mathcal{P}_{\mathfrak{g}}(\rho) = 2^{n-1}/2^n = 1/2$. Since $X_1 \in i\mathfrak{g}$, $\mathcal{P}_{\mathfrak{g}}(O) = \mathrm{Tr}(X_1^2)^2/2^n = 2^n$. Therefore $\mathrm{Var}[\ell] = 2^{n-1}/\dim \mathfrak{k}_n$, and Theorem 1 gives the result. ■

Consequences.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

1. The ℚ-restriction leaves the BP intact: the variance differs from the Haar value only in the ± 1 of the denominator.
2. The gradient variance $\mathrm{Var}[\partial_\theta \ell]$ shows the same $\Theta(2^{-n})$ scaling (Section 6).
3. Any lower bound of the form $\mathrm{Var}[\partial_\theta \ell] \geq \gamma/\mathrm{Poly}(n)$ for generic ℚ-circuits is false.

Escaping the BP requires restricting the DLA *further*, to polynomial dimension inside $\mathfrak{k}_n$.

5. A Trainable ℚ-Structured Family — and Its Classical Simulability

Definition 3 (ℚ-cluster ansatz). The ansatz uses generators

$$\mathcal{G}_{\mathrm{cl}} = \{X_j\}_{j=1}^{n} \cup \{Z_j X_{j+1} Z_{j+2}\}_{j=1}^{n-2}.$$

Every generator has odd weight (1 or 3), so the circuit is ℚ-structured by Proposition 2.

Theorem 3 (polynomial DLA). With $a = \lceil n/2 \rceil$ and $b = \lfloor n/2 \rfloor$,

$$\mathfrak{g}_{\mathrm{cl}} \cong \mathfrak{so}(2a) \oplus \mathfrak{so}(2b), \qquad \dim \mathfrak{g}_{\mathrm{cl}} = a(2a-1) + b(2b-1) \in \Theta(n^2).$$

Proof. Apply the Jordan–Wigner map with X as the on-site operator, with Majorana operators $\gamma_1, \ldots, \gamma_{2n}$ and $X_j \propto i\gamma_{2j-1}\gamma_{2j}$, $Z_j Z_{j+1} \propto i\gamma_{2j}\gamma_{2j+1}$. Then

$$Z_j X_{j+1} Z_{j+2} = -\left(Z_j Z_{j+1}\right) X_{j+1} \left(Z_{j+1} Z_{j+2}\right) \propto i\gamma_{2j}\gamma_{2j+3},$$

so every generator is a quadratic Majorana monomial $i\gamma_r\gamma_s$, and $\mathfrak{g}_{\mathrm{cl}} \subseteq \mathfrak{so}(2n)$. Draw the graph whose edges are the pairs (r, s). Its edges $(2j-1, 2j)$ and $(2j, 2j+3)$ form two paths: one through the Majoranas of the odd-indexed sites and one through those of the even-indexed sites, with $2a$ and $2b$ vertices respectively. Quadratic monomials along a connected tree on m Majoranas generate the full $\mathfrak{so}(m)$, because the bracket of $i\gamma_r\gamma_s$ and $i\gamma_s\gamma_t$ is proportional to $i\gamma_r\gamma_t$. The two path components commute, which gives the direct sum. ■

Theorem 4 (polynomial variance). For the deep ℚ-cluster ansatz with $\rho = |+\rangle\langle+|^{\otimes n}$ and $O = X_1$,

$$\mathrm{Var}_{\boldsymbol{\theta}}[\ell] = \frac{1}{2\lceil n/2 \rceil - 1} \in \Theta(1/n).$$

The formula is exact for $n \geq 5$, where the relevant component $\mathfrak{so}(2a)$ is simple; it is confirmed numerically for $3 \leq n \leq 11$.

Proof. X_1 lies in the component $\mathfrak{so}(2a)$ of the odd sublattice, so only that component contributes in the reductive form of (1). Its Pauli basis contains exactly a X-strings, the on-site X_j for odd j. Hence $\mathcal{P}(\rho) = a/2^n$, $\mathcal{P}(O) = 2^n$, and $\mathrm{Var}[\ell] = a/\big(a(2a-1)\big)$. ■

Theorem 5 (classical simulability). For product input states and observables in $i\mathfrak{g}_{\mathrm{cl}}$ (or in its quadratic-Majorana span), the loss and its gradients for the ℚ-cluster ansatz can be computed classically in $\mathrm{poly}(n)$ time.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

Proof. By Theorem 3, every gate is a free-fermion (matchgate) evolution. Heisenberg-evolved observables in $i\mathfrak{g}_{\mathrm{cl}}$ stay in a $\Theta(n^2)$-dimensional space, which is the setting of g-sim [6] and matchgate simulation [18]. ■

The two results together are an instance of the absence-of-BP / simulability correspondence of Cerezo *et al.* [5]. A ℚ-structure can be *combined* with a small DLA to give trainable circuits, but the known small-DLA mechanisms inside $\mathfrak{k}_n$ are classically tractable. Finding a ℚ-structured family that is trainable *and* not efficiently simulable is left as an open problem.

On gradient evaluation. For $U(\theta) = \exp(\theta\mathbf{u})$ with $\mathbf{u} \in \mathrm{Im}\,\mathbb{Q}$ and $\mathbf{u}^2 = -1$, we have $\partial_\theta U = \mathbf{u}\,U = U\,\mathbf{u}$. This is the standard structure behind the parameter-shift rule [11, 12]. Exact gradients are available for any Pauli-rotation circuit, ℚ-structured or not. A BP concerns the *size* of the gradient relative to shot noise, not how the gradient is computed, so exact gradients do not mitigate BPs.

On adversarial training. A quantum GAN [17] whose generator is a ℚ-structured PQC inherits the generator's DLA. Theorems 2–4 therefore apply to it unchanged: a generic ℚ-generator has a BP, and a ℚ-cluster generator does not but is classically simulable. The earlier version's QGAN immunity claim is withdrawn.

6. Numerical Verification

Methods. The validation code is described in Appendix B. DLA dimensions were computed by exact Lie closure in the Pauli basis. Commutators of Pauli strings are $\pm 2i$ times a Pauli string or zero, so closure under brackets with the generators enumerates a basis of $\mathfrak{g}$. Variances were estimated by exact state-vector simulation with independent uniformly random parameters $\theta_t \in [0, 2\pi]$. The circuits had L layers of all generators in sequence, with $L = 4n$ for the ℚ ansätze and $L = 10n$ for the hardware-efficient baseline. We used $B = 2000$ samples for $n \leq 8$ and $B = 600$ above. The gradient was taken with respect to the middle parameter by the parameter-shift rule. The baseline "HEA" uses generators $\{X_j, Z_j\} \cup \{Z_j Z_{j+1}\}$, which generate $\mathfrak{su}(2^n)$.

Table 1. DLA dimensions from exact Lie closure. $\mathfrak{k}_n$ is the ℚ-symmetry bound of Theorem 1; the cluster column matches $\mathfrak{so}(2\lceil n/2\rceil) \oplus \mathfrak{so}(2\lfloor n/2\rfloor)$ exactly.

n	dim $\mathfrak{su}(2^n)$ (HEA)	dim $\mathfrak{k}_n$ (Thm 1)	ℚ-entangling (computed)	ℚ-cluster (computed)	ℚ-cluster (Thm 3)
2	15	6	6	2	2
3	63	36	36	7	7
4	255	120	120	12	12
5	1023	528	528	21	21
6	4095	2016	2016	30	30
7	16383	8256	8256	43	43

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

n	dim su(2^n) (HEA)	dim $\mathfrak{k}_n$ (Thm 1)	ℚ-entangling (computed)	ℚ-cluster (computed)	ℚ-cluster (Thm 3)
8	65535	32640	32640	56	56
9	262143	131328	131328	73	73
10	—	523776	—	90	90
11	—	2098176	—	111	111
12	—	8386560	—	132	132

Table 2. Loss variance: simulation vs. exact prediction ($\rho = |+\rangle\langle+|^{\otimes n}$, $O = X_1$). Grad. column: simulated variance of $\partial\ell/\partial\theta$ for the middle parameter.

n	HEA sim	Haar $1/(2^n+1)$	ℚ-ent sim	ℚ-ent Thm 2	ℚ-ent grad	ℚ-cluster sim	ℚ-cluster Thm 4	ℚ-cluster grad
3	1.13e-1	1.11e-1	1.13e-1	1.11e-1	6.40e-2	3.33e-1	3.33e-1	2.16e-1
4	6.08e-2	5.88e-2	7.01e-2	6.67e-2	3.30e-2	3.26e-1	3.33e-1	2.20e-1
5	2.90e-2	3.03e-2	3.09e-2	3.03e-2	1.63e-2	2.01e-1	2.00e-1	1.13e-1
6	1.62e-2	1.54e-2	1.61e-2	1.59e-2	8.27e-3	2.08e-1	2.00e-1	1.06e-1
7	8.13e-3	7.75e-3	7.60e-3	7.75e-3	4.18e-3	1.50e-1	1.43e-1	6.87e-2
8	—	3.89e-3	4.11e-3	3.92e-3	2.06e-3	1.41e-1	1.43e-1	6.89e-2
9	—	1.95e-3	2.07e-3	1.95e-3	9.35e-4	1.09e-1	1.11e-1	5.40e-2
10	—	9.76e-4	9.97e-4	9.78e-4	5.73e-4	1.08e-1	1.11e-1	4.06e-2
11	—	4.88e-4	—	4.88e-4	—	9.92e-2	9.09e-2	3.72e-2

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

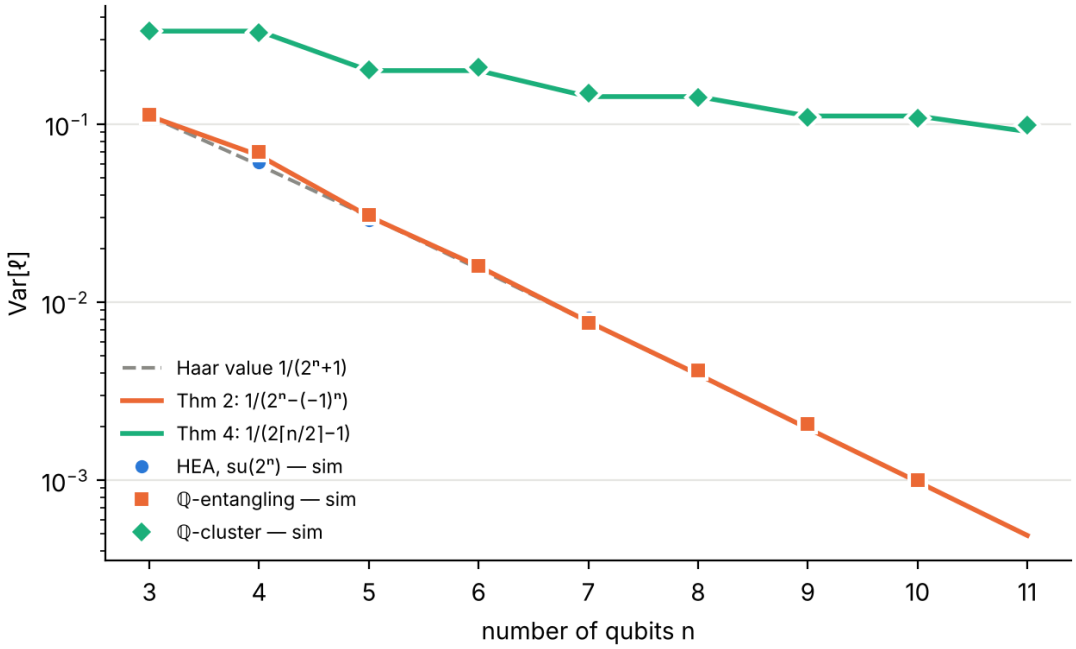


Figure 1. Loss variance versus qubit number (log scale). Markers: simulation; lines: exact predictions from Theorems 2 and 4 and the Haar value. The generic ℚ-ansatz tracks the Haar barren plateau; only the free-fermionic ℚ-cluster ansatz decays polynomially.

Findings. Lie closure reproduces Theorems 1 and 3 exactly at every size computed (Table 1). Across all simulated sizes, the ℚ-entangling variances agree with Theorem 2, and the ℚ-cluster variances agree with Theorem 4 for n ≥ 5, within a maximum relative deviation of 9%, which is consistent with Monte-Carlo error at the sample sizes used. The ℚ-entangling variance falls by about a factor of 2 per added qubit, the same rate as the su(2^n) baseline. The ℚ-cluster variance falls only as 1/n. Gradient variances follow the same scaling as loss variances in every family (Table 2).

7. Conclusion

ℚ-numbers are the quaternions, and their natural n-qubit extension is the antiunitary symmetry $J_n = \sigma_y^{\otimes n} K$. This symmetry confines a circuit's DLA to $\mathfrak{sp}(2^{n-1})$ or $\mathfrak{so}(2^n)$, which is still exponentially large. For a generic entangling ℚ-circuit the loss variance equals the Haar value up to ± 1 in the denominator, so the ℚ-structure is not by itself a barren-plateau mitigation. Polynomial trainability inside $\mathfrak{k}_n$ is possible, as the ℚ-cluster ansatz shows, but the mechanism we found is free-fermionic and classically simulable. The value of the ℚ-framework is as an exact symmetry-design tool: it predicts DLAs and variances in closed form and so diagnoses trainability before any experiment. The open question is whether some ℚ-structured family is both trainable and classically hard.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

Appendix A. Erratum Relative to the Earlier Version

Earlier claim	Status	Correction
$\mathbb{Q}$ is a novel hypercomplex formulation	Incorrect	$\mathbb{Q} \cong \mathbb{H}$; quaternionic QM and quaternion-parameterized qubit gates are established [14–16]
A $\mathbb{Q}$-number represents a pure or mixed state	Incorrect	A unit $\mathbb{Q}$-number is an SU(2) gate; states are orbits (Prop. 1)
Lemmas 1–5	Correct (standard)	Retained; Lemma 3 proof completed
Representation: $\mathfrak{q}$ as placeholder for $\|1\rangle$ with $Z = \alpha + \beta\mathfrak{q}$	Inconsistent	With $\varphi(\mathfrak{q}) = i\sigma_y$, $\varphi(Z)\|0\rangle = \alpha\|0\rangle - \bar{\beta}\|1\rangle$; fixed by $Z = \alpha + \mathfrak{q}\beta$ and $\varphi(\mathfrak{q}) = -i\sigma_y$ (Section 2)
Prop. 1: unique parameterization	Incorrect	Unique only up to a U(1) right action (Hopf fibration)
Prop. 2: derivative of $\exp(\theta\mathfrak{q})$	Correct, trivial	Retained as remark; equivalent to parameter-shift structure
Prop. 3: $\mathbb{Q}\otimes^n$ quaternionic, invariant under Sp(n)	Incorrect	Quaternionic only for odd n; Sp(n) acts on $\mathbb{C}^{2n}$, not on the 2^n-dimensional register; correct group in Thm. 1
Prop. 4: cross term $\mathrm{Tr}((iZ)^\dagger H(\mathfrak{q}Z)) = 0$	False	Counterexample: $Z = 1$, $H = \sigma_x$ gives $2i$
Prop. 5: dimension inequality	Correct	But the gap is only a factor ≈ 2 (Cor. 1)
Thm. 1: DLA polynomial	False	$\dim = (4^n - (-2)^n)/2$, exponential (Thm. 1)
Thm. 2: $\mathrm{Var} \geq \gamma/\mathrm{Poly}(n)$	False	$\mathrm{Var} = 1/(2^n - (-1)^n)$ (Thm. 2)
Thm. 3: non-Haar $\Rightarrow \partial C \neq 0$	Non sequitur	Withdrawn
Thm. 4: gradient without finite differences	Not novel	Parameter-shift rule [11, 12]
Thm. 5: QGAN immunity	Unsupported	Withdrawn; see Section 5
References [1]–[5]	Three of five unverifiable or mis-attributed	Replaced with the verified list below

Appendix B. Validation Code

Every dimension and variance claim in this paper can be checked independently with the open validation code released with it under the MIT License:

Repository: **https://github.com/teknoquants/ALGEBRAIC-FORMULATION-OF--NUMBERS-IN-QUANTUM-SYSTEMS-AND-MITIGATION-OF-THE-BARREN-PLATEAU-PHENOMENON**

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

The code is plain Python 3 with NumPy and Matplotlib only, about 200 lines, and is organized as follows.

- `dla.py` computes dynamical Lie algebras by exact Lie closure in the Pauli basis. It reproduces Theorem 1 (the $\mathbb{Q}$-symmetry bound $\dim \mathfrak{k}_n = (4^n - (-2)^n)/2$), Proposition 3 (saturation by the entangling $\mathbb{Q}$-ansatz) and Theorem 3 (the $\mathbb{Q}$-cluster DLA), i.e. Table 1.
- `sim.py` runs batched state-vector simulations of deep random circuits and estimates the loss variance and the parameter-shift gradient variance. It reproduces Theorems 2 and 4, i.e. Table 2 and Figure 1.
- `pred.py` evaluates the exact variance predictions from the reductive decomposition of the DLA.

Pauli strings are stored as pairs of bit masks (x, z). Two strings anticommute if and only if $|x_1 \wedge z_2| + |z_1 \wedge x_2|$ is odd, and their product has masks $(x_1 \oplus x_2, z_1 \oplus z_2)$. Because the bracket of two Pauli strings is either zero or $\pm 2i$ times a Pauli string, closing the generator set under brackets with the generators enumerates a basis of $\mathfrak{g}$. The core of the closure routine is:

```
def anticommute(a, b):              # a = (x_mask, z_mask)
    n = bin(a[0] & b[1]).count("1") + bin(a[1] & b[0]).count("1")
    return n & 1

def closure(gens):
    basis, queue = set(gens), list(gens)
    while queue:
        p = queue.pop()
        for g in gens:
            if anticommute(p, g):
                r = (p[0] ^ g[0], p[1] ^ g[1])   # [p, g] ∝ p·g
                if r not in basis:
                    basis.add(r); queue.append(r)
    return basis                    # dim g = len(basis)
```

To reproduce the results, run `python dla.py` (Table 1, a few seconds) and `python sim.py` (Table 2, about 15 minutes on a laptop for $n \leq 11$). The random seed is fixed at 7. Pauli rotations act on the state vector through an index permutation and a sign vector, batched over parameter samples.

References

[1] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, “Barren plateaus in quantum neural network training landscapes,” *Nature Communications*, vol. 9, art. 4812, 2018.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

[2] M. Larocca, P. Czarnik, K. Sharma, G. Muraleedharan, P. J. Coles, and M. Cerezo, "Diagnosing barren plateaus with tools from quantum optimal control," *Quantum*, vol. 6, p. 824, 2022.

[3] M. Ragone, B. N. Bakalov, F. Sauvage, A. F. Kemper, C. Ortiz Marrero, M. Larocca, and M. Cerezo, "A Lie algebraic theory of barren plateaus for deep parameterized quantum circuits," *Nature Communications*, 2024, doi:10.1038/s41467-024-49909-3.

[4] E. Fontana, D. Herman, S. Chakrabarti, N. Kumar, R. Yalovetzky, J. Heredge, S. H. Sureshbabu, and M. Pistoia, "Characterizing barren plateaus in quantum ansätze with the adjoint representation," *Nature Communications*, 2024, doi:10.1038/s41467-024-49910-w.

[5] M. Cerezo, M. Larocca, D. García-Martín, N. L. Diaz, P. Braccia, E. Fontana, M. S. Rudolph, P. Bermejo, A. Ijaz, S. Thanasilp, E. R. Anschuetz, and Z. Holmes, "Does provable absence of barren plateaus imply classical simulability?," *Nature Communications*, vol. 16, art. 7907, 2025.

[6] M. L. Goh, M. Larocca, L. Cincio, M. Cerezo, and F. Sauvage, "Lie-algebraic classical simulations for quantum computing," arXiv:2308.01432, 2023.

[7] R. Wiersema, E. Kökcü, A. F. Kemper, and B. N. Bakalov, "Classification of dynamical Lie algebras of 2-local spin systems on linear, circular and fully connected topologies," *npj Quantum Information*, vol. 10, art. 110, 2024.

[8] E. R. Anschuetz and B. T. Kiani, "Quantum variational algorithms are swamped with traps," *Nature Communications*, vol. 13, art. 7760, 2022.

[9] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, "Cost function dependent barren plateaus in shallow parametrized quantum circuits," *Nature Communications*, vol. 12, art. 1791, 2021.

[10] Z. Holmes, K. Sharma, M. Cerezo, and P. J. Coles, "Connecting ansatz expressibility to gradient magnitudes and barren plateaus," *PRX Quantum*, vol. 3, 010313, 2022.

[11] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, "Quantum circuit learning," *Physical Review A*, vol. 98, 032309, 2018.

[12] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, "Evaluating analytic gradients on quantum hardware," *Physical Review A*, vol. 99, 032331, 2019.

[13] F. J. Dyson, "The threefold way. Algebraic structure of symmetry groups and ensembles in quantum mechanics," *Journal of Mathematical Physics*, vol. 3, pp. 1199–1215, 1962.

[14] S. L. Adler, *Quaternionic Quantum Mechanics and Quantum Fields*. Oxford University Press, 1995.

[15] K. Endo, Y. Sato, R. Raymond, K. Wada, N. Yamamoto, and H. C. Watanabe, "Optimal parameter configurations for sequential optimization of variational quantum eigensolver," arXiv:2303.07082, 2023.

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License

[16] J. H. Rúa Muñoz, J. E. Mahecha Gómez, and S. Pineda Montoya, "Quaternionic and octonionic frameworks for quantum computation: mathematical structures, models, and fundamental limitations," *Quantum Reports*, vol. 7, art. 55, 2025.

[17] S. Lloyd and C. Weedbrook, "Quantum generative adversarial learning," *Physical Review Letters*, vol. 121, 040502, 2018.

[18] R. Jozsa and A. Miyake, "Matchgates and classical simulation of quantum circuits," *Proceedings of the Royal Society A*, vol. 464, pp. 3089–3106, 2008.

[19] K. Wada, R. Raymond, Y. Sato, and H. C. Watanabe, "Sequential optimal selections of single-qubit gates in parameterized quantum circuits," *Quantum Science and Technology*, vol. 9, 035030, 2024.

[20] R. Mosseri and R. Dandoloff, "Geometry of entangled states, Bloch spheres and Hopf fibrations," arXiv:quant-ph/0108137, 2001.

Licence

MIT License

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V.

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.